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Key Points:

- The stratigraphic age of the Jiaguan Formation has the range of 128.0–83.6 Ma
- There were two stages of tectonic uplift in Longmen Mountain during the Cretaceous-Early Paleogene (at 128.0–124.1 and 68.8–64.1 Ma)
- The stress field generated by the subduction and closure of the Bangong-Nujiang Ocean had already affected to the eastern Tibet at ~128 Ma

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Upper Cretaceous–Paleocene Magnetostratigraphy of the Longmenshan Foreland Basin and Its Implications for the Tectonic Deformation of the Longmenshan Fault Zone

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Abstract Knowledge of the history of the tectonic deformation of the eastern edge of Tibet is important for understanding the evolution of the entire Tibetan Plateau; however, the corresponding tectonic deformation process remains controversial. We conducted a detailed rock magnetic and magnetostratigraphic study of the Cretaceous Jiaguan and Guankou Formations (the Jinmenguan section) within the Longmenshan foreland basin on the eastern edge of Tibetan Plateau. The high-temperature remanence components separated after thermal demagnetization passed the A-class reversal test, Bootstrap Cartesian coordinate test, and fold test, at the 95% confidence level. The resulting magnetostratigraphy constrains the age range of the Jinmenguan section to 128.0–64.1 Ma, and that of the Jiaguan Formation to 128.0–83.6 Ma. The resulting chronology identifies two phases of rapid sedimentation (128.0–124.1, 68.8–64.1 Ma) that constrain two phases of tectonic uplift in the Longmenshan Fault Zone on the eastern margin of Tibet during the Cretaceous–early Cenozoic. Our findings suggest that the stress field generated by the subduction and closure of the Bangong-Nujiang Ocean had already been transferred to the eastern edge of the Tibet in the Cretaceous (at ~128 Ma).

Plain Language Summary Recent studies have suggested that a significant crustal thickness had already developed on the eastern edge of Tibet before the India-Asia collision, and that the eastern extent of the Tibetan Plateau formed by the India-Asia collision was controlled by the tectonic boundary (the Longmenshan Fault Zone [LMSFZ]) of the eastern edge of Tibet that had already formed during the Cretaceous. The tectonic deformation of the eastern edge of Tibet during the Cretaceous-Early Cenozoic provides an important window for understanding the evolution of the entire Tibetan Plateau. The Longmenshan foreland basin is located on the eastern edge of the Tibet. The stratigraphy within the Longmenshan foreland basin can provide a detailed record of the tectonic deformation process of the eastern edge of Tibet. To further constrain the tectonic deformation process of the LMSFZ on the eastern boundary of Tibet, we conducted a detailed rock magnetic and magnetostratigraphic study within the Longmenshan foreland basin. Our results suggest the following. (a) The LMSFZ experienced two stages of uplift during the Cretaceous-Early Cenozoic (at 128.0–124.1 and 68.8–64.1 Ma). (b) The stress field generated by the subduction and closure of the Bangong-Nujiang Ocean had already affected the eastern edge of the Tibet in the Early Cretaceous.

1. Introduction

Knowledge of the history of the tectonic deformation of the eastern edge of Tibet is important for understanding the evolution of the entire Tibetan Plateau. Different from our early understanding, recent research indicates that the Longmenshan Fault Zone (LMSFZ) was strongly deformed in the late Mesozoic (e.g., Y. Li et al., 2016; Xu et al., 2007; Z. Xue et al., 2017; Z. H. Xue et al., 2022; Wallis et al., 2003), and the Longmen Mountains crust was already thick at the time of the India-Asia collision (e.g., Wallis et al., 2003; Z. Xue et al., 2017; Z. H. Xue et al., 2022). Wallis et al. (2003) suggested that the current crustal thickness of the Longmen Mountains was formed mainly before the India-Asia collision. Tectonic deformation analysis indicates that the Longmen Mountain crust thickened by at least 20 km during the late Mesozoic (Z. Xue et al., 2017; Z. H. Xue et al., 2022). Xu et al. (2007) suggested that the extent of the Tibetan Plateau formed by the India-Asia collision was controlled by the tectonic boundary between the eastern margin (LMSFZ) and the northern margin of Tibet that had already formed during the Cretaceous. However, the timing of the activity and crustal thickening in the LMSFZ during the Cretaceous-early Cenozoic is not precisely constrained. Although several thermochronological studies have

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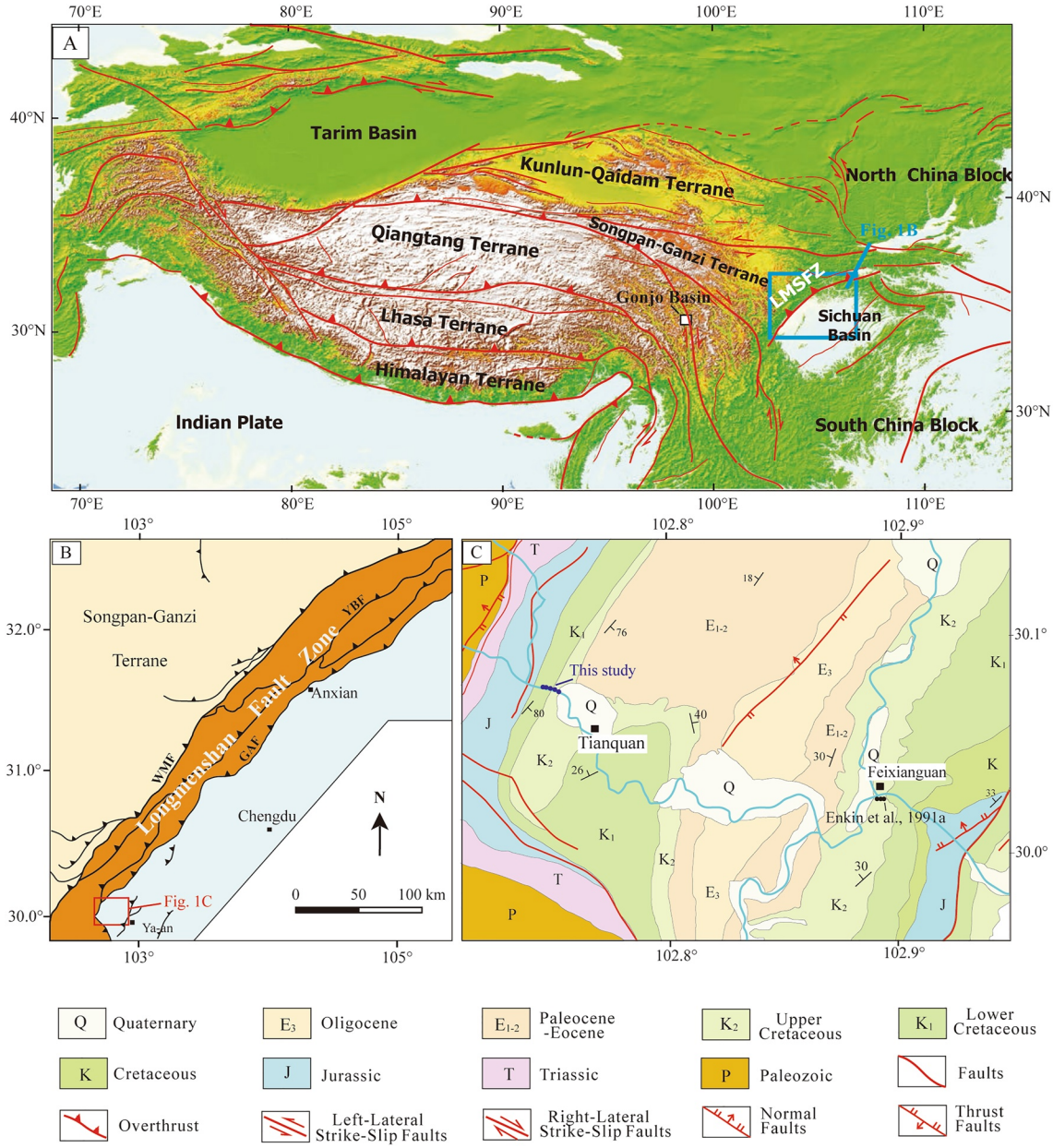


Figure 1. (a) Schematic tectonic geological map of the Sichuan Basin. (b) Structural diagram of the Longmenshan Fault Zone. WMF–Wenchuan-Maoxian Fault; YBF–Yinxiu-Beichuan Fault; GAF–Guanxian-Anxian Fault. (c) Tectonic map of the southwestern edge of the Sichuan Basin and the locations of the sampling sections (modified from Z. W. Li et al., 2012 and Enkin et al., 1991). LMSFZ–Longmenshan Fault Zone.

constrained the timing of the rapid cooling of the LMSFZ, this timing has a broad range and is controversial (Arne et al., 1997; Z. W. Li et al., 2012; Kirby et al., 2002; Z. H. Xue et al., 2022). For example, Arne et al. (1997) suggested that the LMSFZ underwent rapid cooling during ~130–120 Ma; Z. W. Li et al. (2012) suggested that it underwent rapid cooling at ~110 Ma; and Kirby et al. (2002) suggested that it underwent slow cooling during ~130–22 Ma. Therefore, the processes involved in the tectonic deformation of the LMSFZ on the eastern edge of Tibet during the Cretaceous-early Cenozoic require further study.

The Longmenshan foreland basin is located on the eastern edge of the LMSFZ, on the eastern boundary of the Tibetan Plateau (Figure 1a). A complete Cretaceous–Paleogene stratigraphy is developed within the Longmenshan foreland basin, and provenance analysis indicated that during the Cretaceous and Cenozoic the sediment source within the basin was mainly the Longmen Mountains (Y. Li et al., 2016; Peng et al., 2000; Richardson

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et al., 2008). Previous studies have shown that the sediment accumulation rate (SAR) in terrestrial basins depends on the denudation rate of the source area (Hovius, 1998). The main factors affecting the denudation rate are climate and tectonics (which controls the topography) (Chang et al., 2015; Charreau et al., 2006; Molnar & England, 1990). In the absence of significant climate change, structural uplift will lead to an increase in the denudation rate (Chang et al., 2015; Charreau et al., 2006; Hovius et al., 1998; Molnar & England, 1990). During the Cretaceous–early Cenozoic, the LMSFZ was the main provenance area of the Longmenshan foreland basin (Y. Li et al., 2016; Peng et al., 2000; Richardson et al., 2008). Therefore, an increase in the denudation rate of the LMSFZ would have led to an increase of SAR in the Longmenshan foreland basin.

Magnetostratigraphy provides a continuous and precise chronological framework for constraining changes in the SAR in the Longmenshan foreland basin. Based on the SAR in the basin, the timing of the activity of the LMSFZ, on the eastern edge of the Tibetan Plateau, can be precisely constrained. Therefore, the stratigraphy within the Longmenshan foreland basin can potentially provide a detailed record of the tectonic deformation process of the LMSFZ on the eastern edge of Tibet, and magnetostratigraphic studies of the Cretaceous–Paleogene strata in the basin can be used to determine the evolution of the basin via the provision of a precise chronostratigraphic framework for sediment accumulation. To constrain the tectonic deformation process and dynamical mechanism of the LMSFZ on the eastern edge of Tibet, we conducted a high-resolution magnetostratigraphic study of the Early Cretaceous–Paleogene sediments of the Longmenshan foreland basin.

2. Regional Geology and Sampling

The Tibetan Plateau is composed of several blocks—the Kunlun-Qaidam terrane, Songpan-Ganzi terrane, Qiangtang terrane, Lhasa terrane, and Himalayan terrane, from north to south. These blocks collided successively to form the structure of the modern Tibetan Plateau. The LMSFZ, located at the junction of the Songpan-Ganzi terrane and Yangtze terrane, forms the eastern boundary of the Tibetan Plateau and the western boundary of the Sichuan Basin (Figure 1a). The LMSFZ has formed a series of mountains (the Longmen Mountains) during tectonic evolution, and the boundary of the Longmen Mountain is almost the same as that of the LMSFZ. Therefore, previous studies have also named the LMSFZ the Longmen Mountains (Dirks et al., 1994; Jia et al., 2006). The Longmen Mountains are north-east trending, ~600 km long (Xu et al., 2007) and 50–70 km wide (Burchfiel et al., 1995; Y. Li et al., 2016). The elevation of the Longmen Mountains increases rapidly from 500 to 600 m in the eastern Sichuan Basin to over 6,000 m (Burchfiel et al., 1995; Kirby et al., 2002). The Sichuan Basin is at the eastern end of the Longmen Mountains, which has a relatively low relief compared to the Longmen Mountains (Burchfiel et al., 1995). In terms of scale the Longmen Mountains on the eastern margin of the Tibetan Plateau rivals any other margin of the Tibetan Plateau (Burchfiel et al., 1995).

The LMSFZ mainly comprises three large fractures—the Wenchuan-Maoxian Fault, Yingxiu-Beichuan Fault, and Guanxian-Anxian Fault, from west to east (Figure 1b). In between the Wenchuan-Maoxian Fault and the Yingxiu-Beichuan Fault the lithology is mainly pre-Sinian, Silurian, and Devonian shallow metamorphic rocks, and a pre-Sinian complex; and in between the Yingxiu-Beichuan Fault and the Guanxian-Anxian Fault the lithology is mainly unmetamorphosed Paleozoic and Triassic rocks. The ancient LMSFZ was formed by the convergence of the Songpan-Ganzi terrane and Yangtze terrane in the Late Triassic (Burchfiel et al., 1995; Roger et al., 2011; Tian et al., 2016). From the Early Cretaceous, the LMSFZ underwent rapid cooling and denudation (e.g., Arne et al., 1997; Z. W. Li et al., 2012; Liu et al., 2001) and NW-SE-oriented tectonic extrusion (Dirks et al., 1994; Y. Li et al., 2016; Xu et al., 2007; Z. Xue et al., 2017; Z. H. Xue et al., 2022). Later, from the end of the Late Cretaceous to the early Cenozoic, strong tectonic deformation of the LMSFZ occurred due to the collision of India and Asia (Houseman & England, 1993; Tapponnier et al., 2001; Tian et al., 2016).

The Longmenshan foreland basin is located within the southwestern part of the Sichuan basin (Figure 1a). Within the Longmenshan foreland basin, a complete sequence of Cretaceous–Paleogene strata is developed, with the thickness of the Cretaceous strata exceeding 1,000 m. From bottom to top, the Cretaceous strata comprise the Tianmashan Formation (K_1t), Jiaguan Formation (K_1j), and Guankou Formation (K_2g); and the Paleogene strata comprise the Mingshan Formation (E_{1-2}m) and Lushan Formation (E_3l). The sediments of the Jiaguan Formation are of fluvial origin, and the sediments of the Guankou Formation are of lacustrine-fluvial origin (BGMRS, 1991). The lithology of the Jiaguan Formation is mainly brownish-red and purplish-red calcareous cemented medium–fine sandstone, siltstone, and thinly laminated mudstone, with a stratigraphic thickness of ~500 m. In the Jiaguan Formation, large cross beddings and basal scouring structures are developed. The base of

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the Jiaguan Formation consists of a ~5-m-thick conglomerate which defines a clear boundary with the underlying Tianmashan Formation. The lithology of the Guankou Formation is mainly brownish-red and purple siltstone, and pelitic siltstone, with several layers of light gray, thin marlstone, gypsum, and glauberite. At the base of the Guankou Formation, an 11-m-thick conglomerate (interbedded with sandstone) is developed, with a conformable contact with the underlying Jiaguan Formation.

The sampling site (102°44′12″E, 30°04′20″N) is located to the west of Jinmenguan, in Tianquan County, Sichuan Province (Figure 1c). Sampling was conducted in 2013. The stratigraphic outcrops in the sampled section are complete, continuously deposited, and weakly weathered. The samples obtained in this study are from the Cretaceous Jiaguan and lower Guankou Formations. The lithology in the Jinmenguan section is relatively uniform, with only small-scale conglomerate deposits at the base (~10-m-thick, 0–10 m within the section), middle (~9-m-thick, 491–500 m within the section) and top (~14-m-thick, 710–724 m within the section). The lithology of the remaining parts of the Jinmenguan section is mainly purplish-red and brownish-red muddy siltstone and fine sandstone, with a few layers of medium–coarse sandstone and thinly laminated mudstone (a few of the mudstone and siltstone samples are gray in color). The gravels in the conglomerate layer are well rounded and the lithologies of most of the gravels are strongly deformed metamorphic and volcanic rocks with several layers of sandstone (medium and coarse-grained), with ~0.3–1 m-thick interbeds within the conglomerate layers. The sampling interval is 0.8–1.5 m (one sample was collected from each layer), and a total of 595 paleomagnetic samples were collected from the 724.1-m-thick section. Sampling was conducted using a portable gasoline-powered drill and the samples were oriented with a magnetic compass (Figure 2). Considering that the study area is tectonically complex and stratigraphic comparison is difficult, and to maximize the reliability of the results, we only sampled one section with a complete stratigraphic outcrop (the strata in the sampling section are continuously deposited, completely exposed, and have not undergone structural damage) (Figure 2). The sampling stratum is only located within one limb of the fold. The samples were cut into standard paleomagnetic specimens with a diameter of 2.54 cm and a height of 2.3 cm.

3. Results

3.1. Rock Magnetism

To determine the magnetic mineralogy of the samples, eight representative samples were selected for measurements of hysteresis loops, isothermal remanent magnetization (IRM) acquisition and backfield demagnetization, and thermal demagnetization of a three-component IRM (Lowrie, 1990). Representative samples were selected based on lithology and color. JM7-3 is a brownish-red muddy siltstone, JM14-5 is a purplish-red fine sandstone, JM22-3 and JM37-6 are brownish-red fine sandstones, JM36-7 is a purplish-red medium sandstone, JM41-1 is a purplish-red coarse sandstone, JM46-5 is a brownish-red muddy siltstone, and JM50-6 is a gray siltstone mudstone. All the experiments were conducted using an 8,600 Series Vibrating Sample Magnetometer. For the thermal demagnetization of a three-component IRM, DC fields of 2.45, 0.4, and 0.12 T were successively applied to the sample *Z*-, *Y*- and *X*-axes, respectively, prior to stepwise thermal demagnetization. The temperature intervals were the same as those used for stepwise thermal demagnetization of the NRM (Section 3.2).

Except for specimen JM46-5, the hysteresis loops of the representative samples are all wasp-waisted (Figures 3a–3c, 3g, and 3i), likely due to the co-occurrence of magnetite and hematite (Tauxe et al., 1996). The hysteresis loop for specimen JM46-5 is typical of that of hematite (Tauxe et al., 1996), which suggests that the proportion of hematite in this sample is higher than in other representative samples. The IRM acquisition curves of all six representative samples show that the remanence did not approach saturation until ~2 T (Figures 3d–3f and 3j–3l), indicating the presence of a high-coercivity magnetic phase, which is likely to be hematite.

The results of saturation isothermal remanent magnetization (SIRM) component analysis (Figure 4) reveal that all six samples have three magnetization components. The ranges of $B_{1/2}$ (the applied DC field at which half of the SIRM is acquired) for the six representative samples are 537.0–851.1 mT for component 1, 66.1–177.8 mT for component 2, and 7.9–20.9 mT for component 3 (Figure 4a). Additionally, the results of three-axis thermal demagnetization for the six representative samples showed that the remanent magnetic intensity of all six representative samples decreased to zero at ~680°C (Figure 4b), which indicates that their magnetic mineralogy is similar.

In the results for sample JM46-5, the $B_{1/2}$ of component 1 is 537.0 mT (contributing 87% of the SIRM). The thermal demagnetization of a three-component IRM for this sample shows that the soft, intermediate and hard

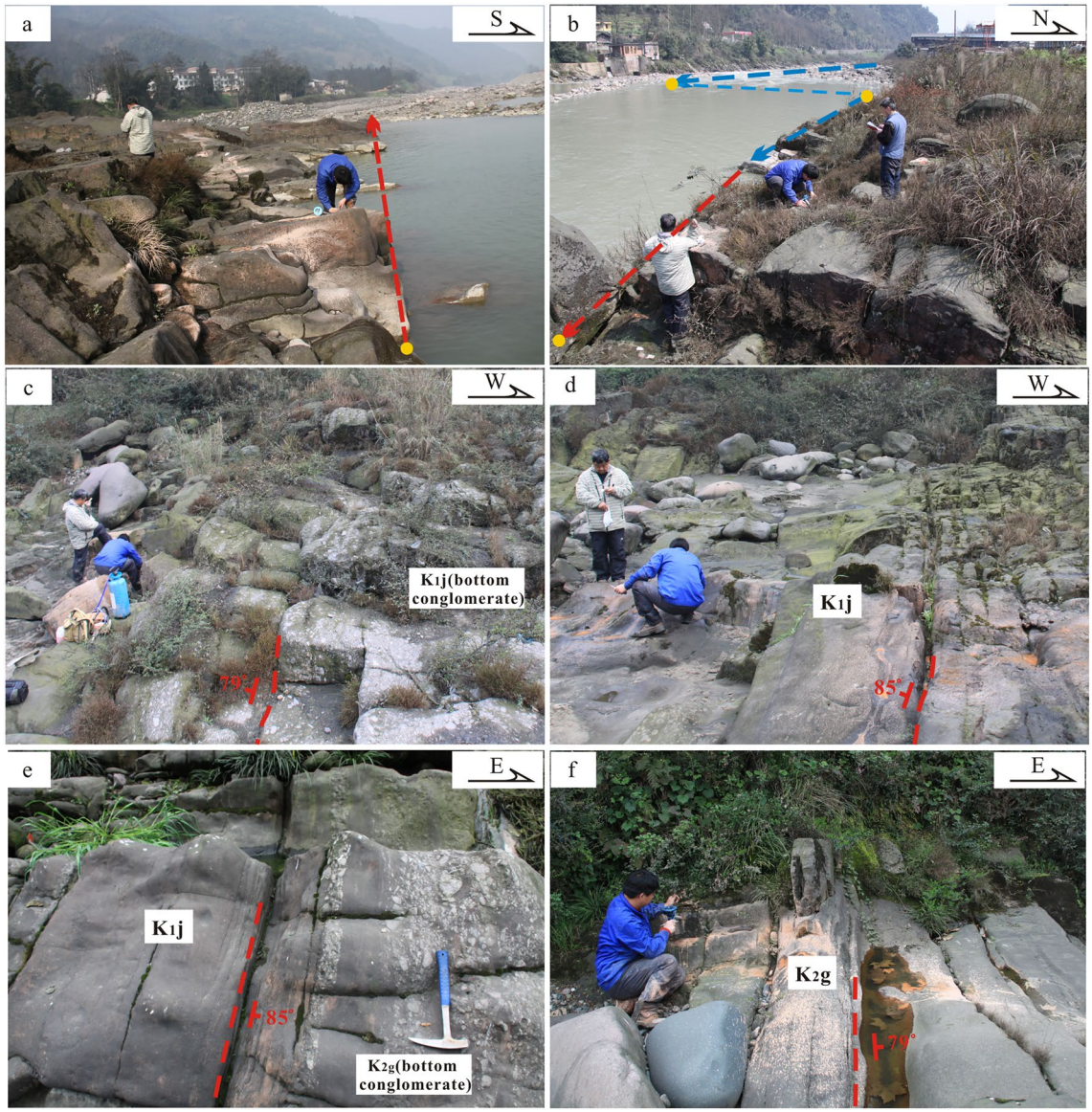


Figure 2. Photos showing field sampling. K_1j = Early Cretaceous Jiaguan Formation, and K_2g = Late Cretaceous Guankou Formation. The blue line in (b) represents the paleomagnetic sampling pathway in the Jiaguan Formation, and the red lines in (a) and (b) represent the sampling pathway in the Guankou Formation. The arrow direction represents the decreasing age of a formation. (c) Conglomerate at the base of the Jiaguan Formation, defining the boundary with the underlying Tianmashan Formation. (d) Middle part of the Jiaguan Formation. (e) Boundary between the Jiaguan Formation and the Guankou Formation (the basal conglomerate in the Guankou Formation). (f) Middle part of the Guankou Formation.

magnetic components decrease rapidly to zero at 680°C, which is the unblocking temperature of hematite, indicating that component 1 is hematite. $B_{1/2}$ of component 2 is ~100 mT (contributing 12% of the SIRM). In the thermal demagnetization of a three-component IRM, the soft component of samples JM7-3 and JM14-5 decreases rapidly at 580°C, corresponding to the Curie temperature of magnetite.

In samples JM14-5, JM36-7, and JM46-5, component 3 only contributes ~1% of the SIRM and it cannot be reliably characterized. In sample JM50-6, $B_{1/2}$ of component 3 is 20.9 mT (7% of the SIRM), and the results of the thermal demagnetization of a three-component IRM show that the curves for all three components have slight inflections at ~350°C, indicating that component 3, with a low $B_{1/2}$, is probably carried by maghemite which begins to be converted to hematite at ~350°C (Dunlop et al., 1997). Overall, the results indicate that the magnetic minerals in the samples are mainly magnetite and hematite, together with a minor amount of maghemite.

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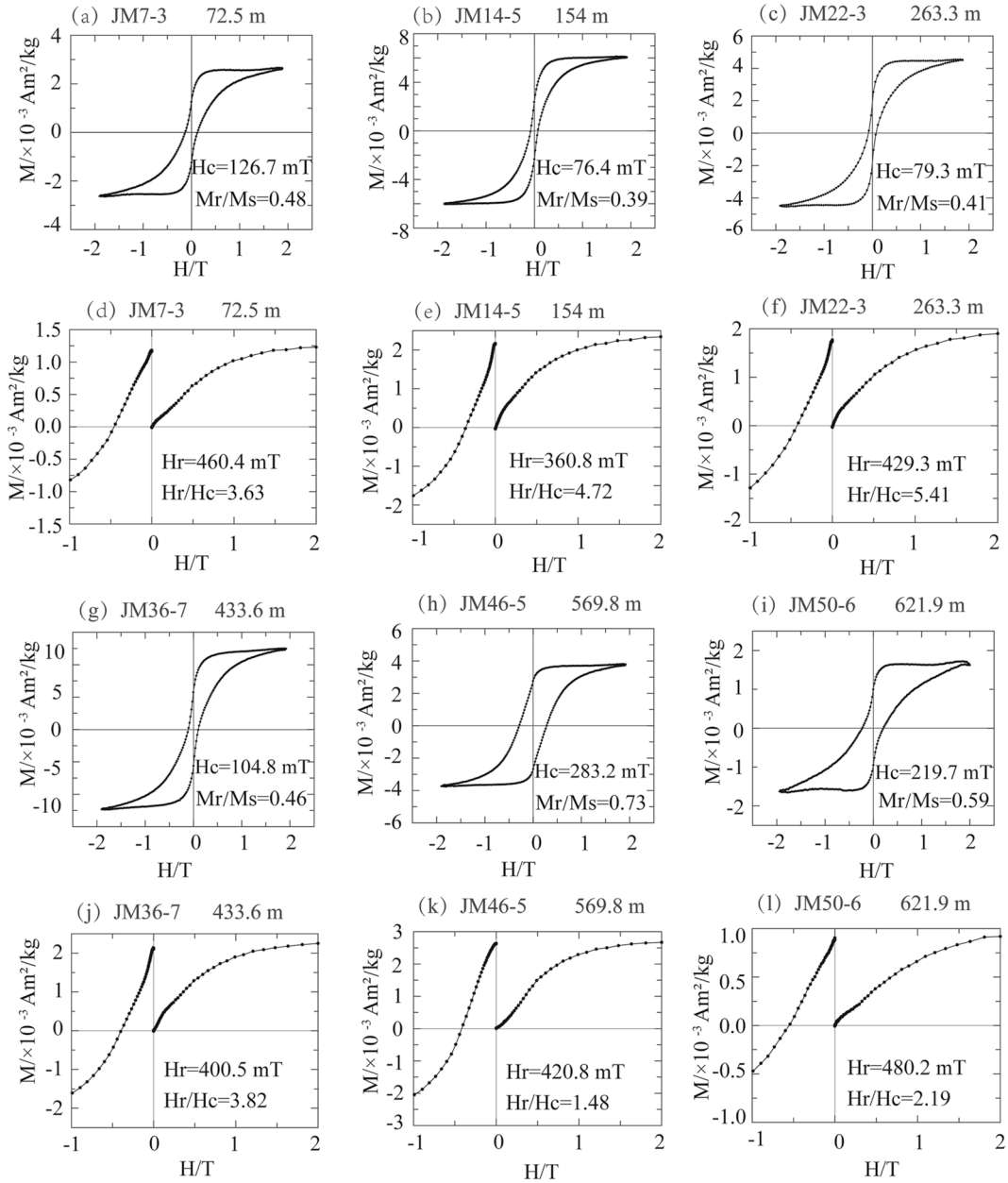


Figure 3. Rock magnetic results for representative samples. (a–c, g–i) Hysteresis loops after paramagnetic slope correction. Isothermal remanent magnetization acquisition and back field demagnetization curves (d–f, j–l).

3.2. Thermal Demagnetization of the NRM

Systematic thermal demagnetization experiments were conducted on all specimens to isolate the characteristic remanent magnetization (ChRM). The measurements were made in a magnetically-shielded room to minimize the influence of the ambient geomagnetic field. The thermal demagnetization steps were 50°C–100°C in the low-temperature range, 30°C–40°C in the intermediate-temperature range, and 10°C–20°C in the high-temperature range. After each step, the remanence of the samples was measured with a 2G-755 superconducting magnetometer. The demagnetization results are presented as Zijderveld diagrams (Zijderveld, 1967). The direction of the ChRM was determined using principal component analysis (Kirschvink, 1980), or by the best-fit great circle method (McFadden & McElhinny, 1988). The averaged high-temperature remanence directions of the samples were obtained using Fisher statistics (Fisher, 1953). All rock magnetic and stepwise thermal demagnetization experiments were completed in the Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of

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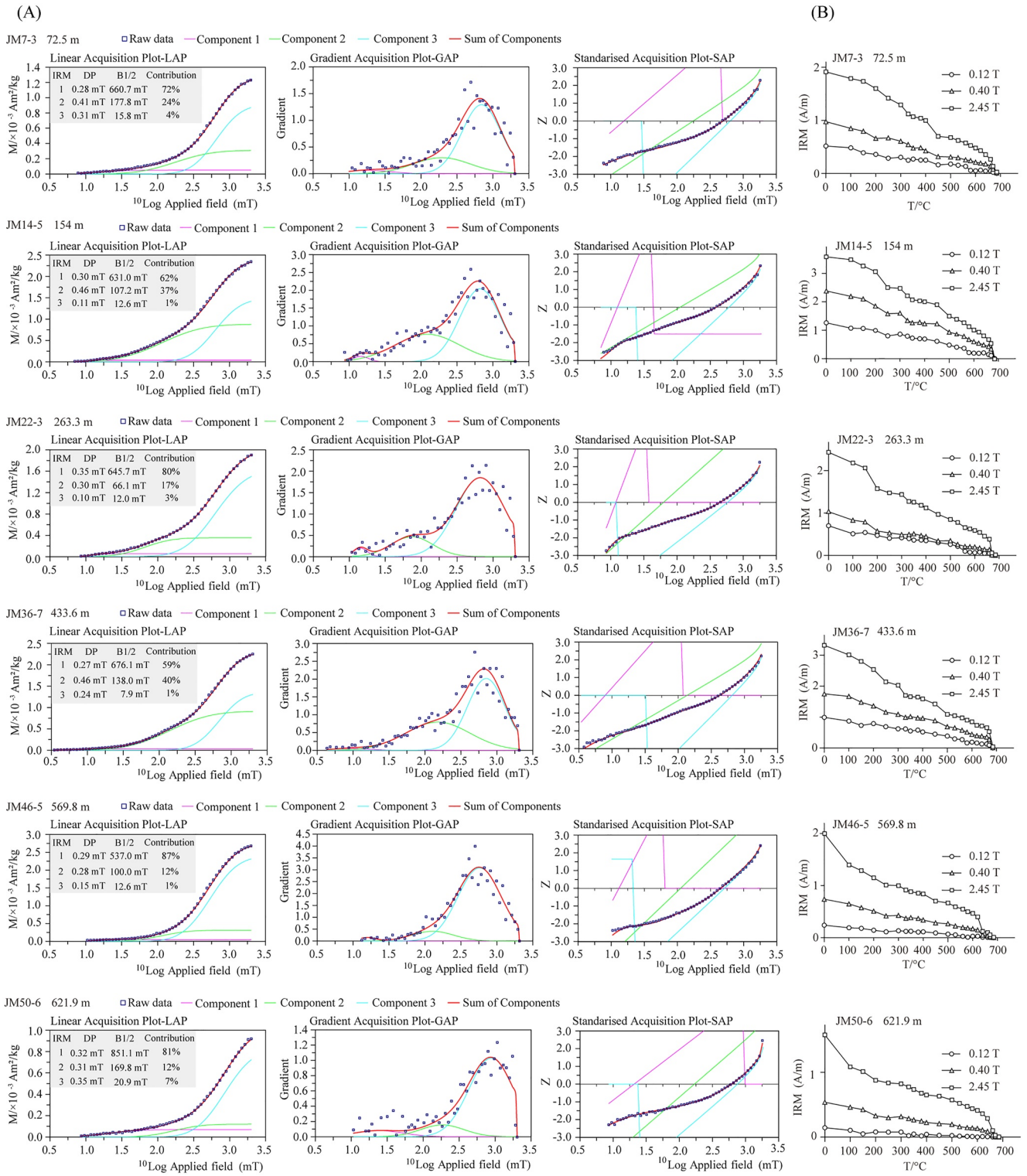


Figure 4. Rock magnetic results for representative samples from the Jinmenguan section. (a) Results of isothermal remanent magnetization (IRM) component analysis (Kruiver et al., 2001). Squares are the measured data, and the components are indicated by lines. $B_{1/2}$ is the field at which one half of the saturation isothermal remanent magnetization is acquired, DP is the dispersion parameter, indicating the width of the distribution. (b) Results of thermal demagnetization of a three-component IRM (Lowrie, 1990).

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Natural Resources, China. Processing and statistical analysis of the paleomagnetic data were conducted using the KIRSCH.EXE and PMSTAT.EXE computer programs written by Randy Enkin (Geological Survey of Canada), the PaleoMac 6.2 computer program written by Cogné (2003), and the online paleomagnetic analysis website (Paleomagnetism.org) developed by Koymans et al. (2016).

The thermal demagnetization results for the Jinmenguan section show that 299 samples have two components with different directions, and 296 samples have a single component (Figure 5). Among them, the low-temperature characteristic component (Dg = 4.0°, Ig = 47.6°, k = 11.4, α_{95} = 2.5°, n = 299 and Ds = 73.1°, Is = 20.4°, k = 10.7, α_{95} = 2.6°, n = 299) was removed at 280°C. The mean direction before tilt correction (Dg = 4.0°, Ig = 47.6°, k = 11.4, α_{95} = 2.5°, n = 299) is close to the geocentric axial dipole direction (GAD) of D = 0.0°, I = 47.6° (Figure 6a). Most of the samples (437 samples) show a linear decay of the magnetization toward the origin within the temperature interval of 460°C–685°C, and thus a high-temperature characteristic magnetization component could be isolated using principal component analysis. In a relatively small number of cases (72 samples), the best-fit great circles method was used to isolate the high-temperature ChRM.

There are several differences in the thermal demagnetization characteristics of the samples with different lithologies and colors. For the purplish-red and brownish-red sandstones, most of the samples show a very gradual decrease in remanent magnetization intensity between 20°C and 580°C, a rapid decrease in remanent magnetization intensity between 610°C and 685°C, and finally a decrease in remanent magnetization intensity to near zero between 660°C and 685°C. Several samples show a significant decrease in remanent magnetization intensity at 580°C. For the purplish-red and brownish-red siltstones, unlike the purplish-red and brownish-red sandstones, some of the samples show a significant decrease in remanent magnetization intensity between 20°C and 280°C. For the gray sandstones and mudstones, the demagnetization characteristics are more complicated because the remanent magnetization intensity shows a significant decrease between 20°C and 280°C, finally decreasing to near zero between 660°C and 685°C for most samples, although the remanences of several samples decrease to near zero between 510°C and 580°C. In 86 samples, a high-temperature characteristic component could not be separated; these samples are mainly medium and coarse-grained sandstones, with a relatively coarse bulk grain size, and gray mudstones and siltstones.

After obtaining the direction of the high-temperature ChRM of each sample, the site-mean direction was obtained using Fisher statistics. A total of 509 samples yielded a high-temperature ChRM with an average direction for the section of Dg = 347.1°, Ig = −1.6°, k = 11.6, α_{95} = 1.9°, n = 509; and Ds = 17.6°, Is = 38.7°, k = 12.0, α_{95} = 1.9°, n = 509 (Figure 6b). This direction is significantly different from the present-day geomagnetic field direction at the sampling section. The separated high-temperature characteristic components have normal and reverse polarities, and the results of an inversion test ($\gamma = 1.2 < \gamma_c = 4.9$) (McFadden & McElhinny, 1990) and Bootstrap Cartesian coordinate test (Tauxe et al., 2010) were both positive at the 95% confidence level, after tilt correction. Thus, we conclude that this component is the primary remanence.

3.3. Magnetostratigraphy

The virtual geomagnetic pole (VGP) of each sample was calculated from the high-temperature ChRM, enabling the magnetic polarity reversal stratigraphy for the Jinmenguan section to be determined. Eight normal polarity events (N1–N8) and nine reverse polarity events (R1–R9) were identified (Figure 7).

Several previous magnetostratigraphic studies were conducted in the Longmenshan foreland basin (Enkin et al., 1991; Shen, 2018; Zhuang et al., 1988). Those of Shen (2018) and Zhuang et al. (1988) constrain the age of the bottom of the Guankou Formation to the mid–Late Cretaceous. Enkin et al. (1991) conducted a high-resolution magnetostratigraphic study in the lower Guankou Formation (sampling thickness of 272 m) and produced a reliable magnetic polarity column that identified two complete normal polarity intervals and three complete reverse polarity intervals. Comparison with the standard geomagnetic polarity time scale (GPTS) resulted in Enkin et al. (1991) proposing three possible correlation options: (a) Option I: C32–C29 (~71–65 Ma, Walker et al., 2018); (b) Option II: M4–M0 (~131–126 Ma, Walker et al., 2018); (c) Option III: M18–M15 (~144–140 Ma, Walker et al., 2018). Enkin et al. (1991) ruled out Option III, and among Option I and Option II they preferred Option II, constraining the ages in the lower part of the Guankou Formation to M4–M0 (~131–126 Ma). From a summary of the results of previous magnetostratigraphic studies, it can be concluded that there are two views of the age framework of the Guankou Formation.

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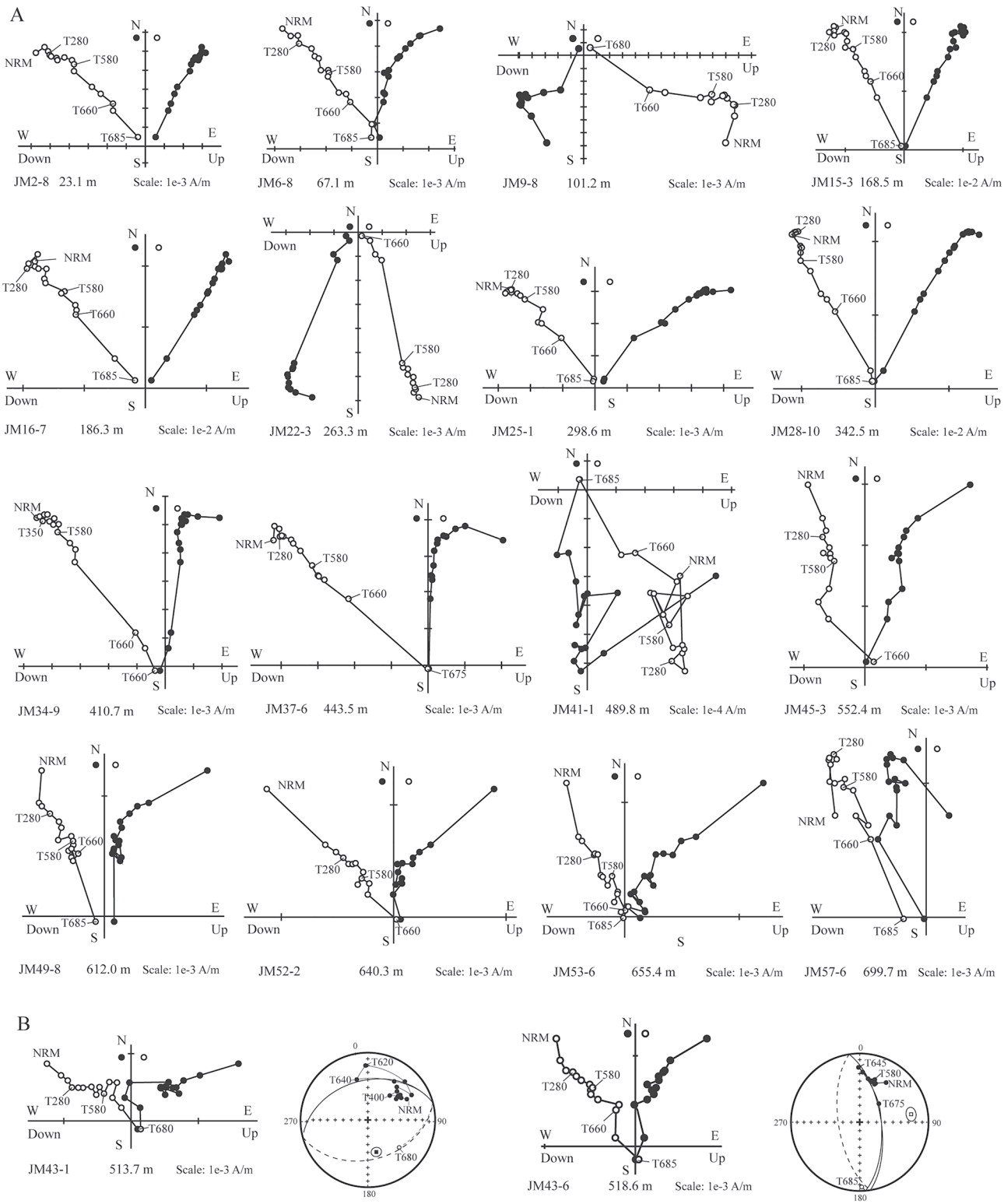


Figure 5. Orthogonal vector diagrams after stepwise thermal demagnetization for representative specimens. (a) High-temperature magnetic components were isolated using principal components analysis (Kirschvink, 1980). (b) High-temperature magnetic components were isolated using the best-fit great circles and intersection of great circles methods (McFadden & McElhinny, 1988). Solid/open symbols indicate horizontal/vertical projections. All results are in geographic coordinates. Numbers adjacent to the decay plots are the heating steps in degrees Celsius. NRM is the natural remanent magnetization.

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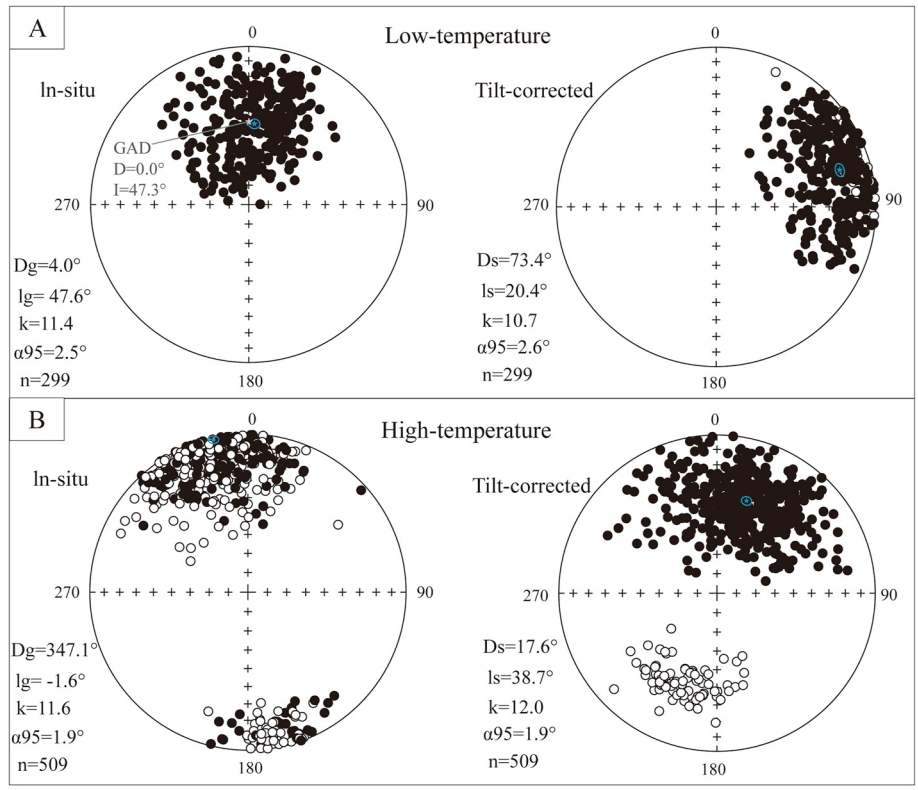


Figure 6. Equal-area projections of the low-temperature remanence component (a) and high-temperature remanence component (b), with in-situ (left) and tilt-corrected (right) paleomagnetic directions for all specimens from the Jinmenguan section. Blue circles enclosing blue stars indicate the 95% confidence level and the mean direction. Solid/open symbols are lower/upper hemisphere projections.

Previous magnetostratigraphic studies provide reference comparisons for our geomagnetic polarity column. According to the results of previous magnetostratigraphic studies, we established two possible comparison schemes. In Scheme 1, our magnetic polarity column corresponds to M1–C28r of the standard GPTS (Walker et al., 2018) (Figure 7), and the basal age of the Guankou Formation is constrained to ~83.6 Ma. Our Scheme 1 resembles that of Shen (2018) and Zhuang et al. (1988), and it is also similar to Option I of Enkin et al. (1991). Considering that Enkin et al. (1991) preferred to correlate the lower part of the Guankou Formation to M3–M0r, we established a second possible comparison scheme. In our Scheme 2, we attempted to correlate polarity intervals R1–R7 of the lower part of the Guankou Formation to the GPTS (Walker et al., 2018) of the Early Cretaceous, but we were unable to establish a reliable correlation (Figures 7f and 7g).

We regard Scheme 1 as reliable for three main reasons: (a) In Scheme 1, our magnetic polarity column correlates perfectly with M1–C28r of the GPTS (Walker et al., 2018). However, a reliable correlation with the GPTS could not be established for Scheme 2. (b) Numerous fossils of the genus *Talicypridea* (the characteristic species are *T. chinensis*, *T. longa*, *T. dedorte*, *T. amoena*, and *T. baizhanensis*) have been discovered in the Upper Jiaguan Formation (the earliest occurrence) and in the lower Guankou Formation strata in the Longmenshan foreland basin (BGMRS, 1991; Wang et al., 2006). Additionally, Cenozoic ostracodes such as *Cypris*, *Cyprinotus*, and *Limnocy* are relatively abundant in the Guankou Formation (BGMRS, 1991; Wang et al., 2006). The genus *Talicypridea* is widely distributed in the Late Cretaceous strata of terrestrial basins in China (e.g., the Donggou Formation in the Junggar Basin of Xinjiang, and the Nanxiong Formation in the Nanxiong Basin of Guangdong) (BGMRS, 1991; Wang et al., 2006), and their age is confined to the Late Cretaceous (BGMRS, 1991; Wang et al., 2006; Yo, 2000). (c) The youngest age obtained from the detrital zircon U–Pb geochronology study conducted by Y. Q. Li, He, et al. (2018) in the central part of the Jiaguan Formation is 130 Ma, representing the maximum age of sedimentation. Therefore, we chose Scheme 1, which correlates the geomagnetic polarity record of the Jinmenguan section to M1–C34n of the GPTS (Walker et al., 2018). In

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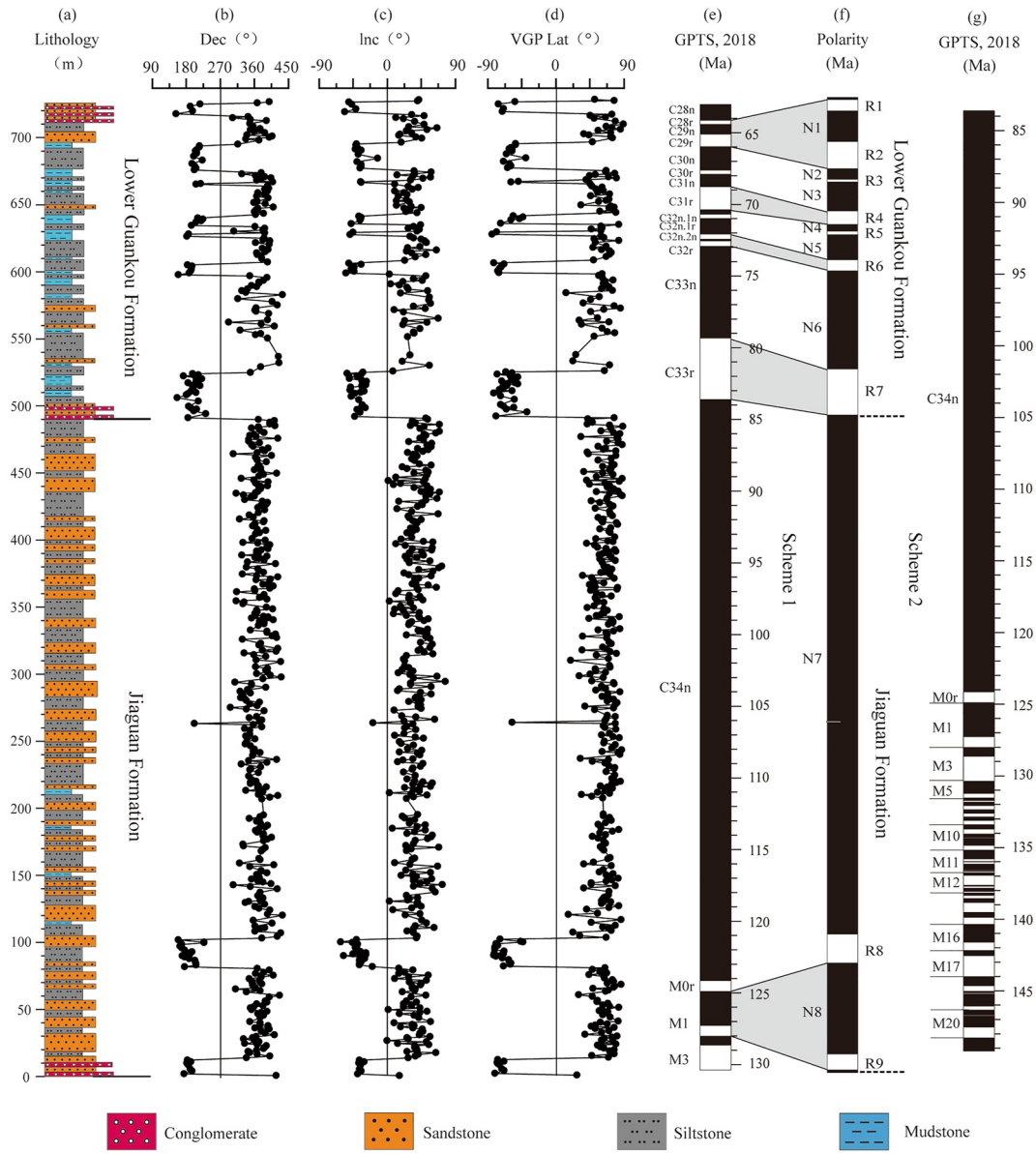


Figure 7. Lithology (a) and observed magnetic directions and polarities (b–g) of the Cenozoic sedimentary sequence in the Sichuan Basin and their correlation with the geomagnetic polarity time scale (Walker et al., 2018). Dec, declination; Inc, inclination; VGP, virtual geomagnetic pole; Lat, latitude of the virtual geomagnetic polarity.

our magnetic polarity column, N7 has the depositional thickness of ~400 m. In Scheme 1, N7 corresponds to C34n of the GPTS (Walker et al., 2018) and the range of variation of SAR in the sampling profile is reasonable (~7–35 m/Ma).

Based on our geomagnetic polarity column, the age of the basal stratigraphy of the Guankou Formation is constrained to 83.6–64.1 Ma (Figure 7). Option I of Enkin et al. (1991) constrains the lower part of the Guankou Formation to ~71–65 Ma (sampling did not reach the bottom of the Guankou Formation), which is similar to our results. The study area is located on the western flank of the fold, while the study area of by Enkin et al. (1991) is located on the eastern flank of the same fold (~10 km from our sampling site) (Figure 1b). Combining the data from the lower Guankou Formation on both limbs of the syncline (contemporaneous paleomagnetic directions), resulted in a positive fold test (McFadden, 1990), at the 95% confidence level, with ξ *in-situ* = 0.37 in geographic coordinates, ξ *tilt-corrected* = 0.10 after tilt correction, and the critical value of ξ *C* = 0.26 at the 95% confidence level.

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4. Discussion

4.1. Stratigraphic Age of the Jiaguan Formation

Precisely constraining the age of the Jiaguan Formation is important for paleoenvironmental and other studies in the Longmenshan foreland basin. Although early studies established a biostratigraphic age for the Jiaguan Formation based on ostracod fauna (BGMRS, 1991; Wang et al., 2006), its precise stratigraphic age is debated. In addition to paleontological studies, several magnetostratigraphic studies have constrained the age of the Jiaguan Formation (Shen, 2018; Zhuang et al., 1988), but the results of Shen (2018) and Zhuang et al. (1988) are not consistent. In the geomagnetic polarity column of Zhuang et al. (1988), the Jiaguan Formation corresponds to the interval of M0r–C34n of the GPTS, and the top of the Jiaguan Formation also corresponds to C34n, which constrains the age of the Jiaguan Formation to the interval of 124.5–83.6 Ma. In the geomagnetic polarity column of Shen (2018), the measured interval of the Jiaguan Formation corresponds to the middle of C34n and the upper boundary corresponds to the top of C34n of the GPTS, which constrains the age of the Jiaguan Formation to the interval of 105.5–83.6 Ma. However, Shen (2018, p. 42) made the following observation: "the sampling profiles are poorly exposed in the range of 1,610–1,760 m, and the samples are mostly fragmented, so the data preserved are sparse; in the range of 1,760–1,860 m, the stratigraphy is mostly missing, and only sparse data of 5 samples were obtained." Additionally, according to Shen (2018), fewer than 30 samples were used to establish a magnetostratigraphy for the depth range of ∼2,200–2,400 m (Jiaguan Formation). Therefore, the sampling density does not meet the requirement for establishing a reliable magnetostratigraphy.

According to our magnetostratigraphy, the Jiaguan Formation is correlated to M1–C34n of the GPTS; the top of the Jiaguan Formation is correlated to the top of C34n; and the base of the Jiaguan Formation is correlated to M1. This correlation reliably constrains the age range of the Jiaguan Formation to 128.0–83.6 Ma. Although our results differ substantially from those of Shen (2018), our magnetostratigraphic age estimate for the Jiaguan Formation is consistent with the biostratigraphic age (Early Cretaceous–Late Cretaceous) (BGMRS, 1991; Wang et al., 2006). Additionally, our magnetostratigraphy is similar to that of Zhuang et al. (1988), but the age of the base of the Jiaguan Formation is 3.5 Myr older than that of Zhuang et al. (1988).

We collected 405 paleomagnetic samples from the 488.6 m–thick strata of the Jiaguan Formation, with sampling intervals mainly at ∼0.8–1.5 m. Zhuang et al. (1988) collected a total of 197 paleomagnetic samples from the Jiaguan and Guankou Formations. Our sampling density is much higher than that of Zhuang et al. (1988), which may explain the difference. Given that the paleomagnetic results presented here passed the reversal and fold tests, our paleomagnetic sampling density is higher, and our magnetostratigraphic age range for the Jiaguan Formation is consistent with the results of a recent biostratigraphic study, we regard our estimate of the age range for Jiaguan Formation to be the most reliable.

4.2. Changes in Sediment Accumulation Rate (SAR)

We used our magnetostratigraphy to calculate the SAR during the interval of 128.0–64.1 Ma (Figure 8). The results reveal two intervals of substantially increased SAR, during 128.0–124.1 Ma (26.2 m/Myr) and 68.8–64.1 Ma (17.9 m/Myr), and one interval of substantially decreased SAR during 124.1–68.8 Ma (9.8 m/Myr) (Figure 8).

It should be noted that the N7 polarity interval (124.1–83.6 Ma) corresponds to the CNS. Since the CNS represents a relatively long time-interval and the SAR only represents the average deposition rate during this interval, some doubt exists about whether the SAR obtained in this study reliably represents the true variations in the SAR. Combining the lithologic features of the N7 polarity interval and the SAR characteristics of the entire stratigraphic column, we infer that the SAR did not change significantly during the CNS, for the following reasons. (a) The lithology of the N7 polarity interval is mainly purplish-red and brownish-red muddy siltstone and fine sandstone interbedded with a few layers of medium sandstone with thicknesses of ∼0.2–0.6 m, and thinly laminated mudstone. Previous studies have shown that an increase in the terrigenous sedimentation rate within the basin usually corresponds to a coarsening of the lithology (Chang et al., 2015; S. H. Li, van Hinsbergen, et al., 2020). Thus, the foregoing lithological summary indicates the absence of any major change in the lithology during the N7 interval. (b) The SAR of N7 (9.5 m/Myr, 124.1–83.6 Ma) is highly consistent with that of the subsequent interval (N4–R7) from 83.6 to 70.8 Ma (8.5–13.7 m/Myr). Therefore, we consider it highly likely that the magnitude of the sedimentation rate variations within the basin was small during the CNS (124.1–83.6 Ma).

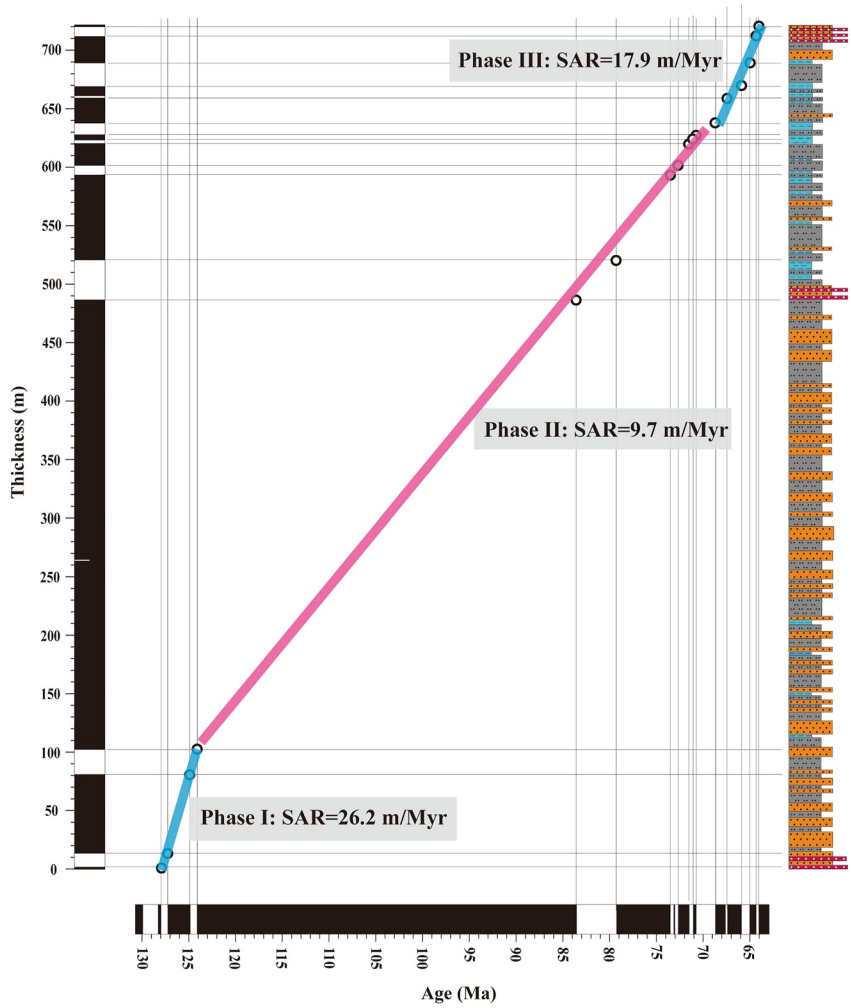


Figure 8. Age-depth plot showing the calculated linear average sediment accumulation rate (SAR) based on the correlations shown in Figure 7. The lithology of the section is shown on the right and the symbols are the same as in Figure 7. The horizontal coordinates are the standard polarity column (Walker et al., 2018), and the vertical coordinates are the magnetic columns of the Jinmenguan section. Phase I (128–124.1 Ma) and Phase III (68.8–64.1 Ma) represent two phases of rapid deposition, and Phase II (124.1–68.8 Ma) represents a phase of slow deposition. The average SAR was 26.2 m/Myr for Phase I, 9.7 m/Myr for Phase II, and 17.9 m/Myr for Phase III.

We then used Option I of Enkin et al. (1991) to calculate the SAR for the eastern flank of the fold (Feixianguan section) during ~71–65 Ma, which yielded values of ~12.5 m/Myr during ~71–69 Ma and ~35 m/Myr during ~69–65 Ma. On the western flank of the fold, we obtained SARs of ~5 m/Myr during ~71–69 Ma and ~15 m/Myr during ~69–64 Ma. Although the SARs vary between the two flanks of the fold (by a factor of ~3), they both increased substantially at ~69–65 Ma. BGMRS (1991) reported that the sediment thickness of the Guankou Formation is ~410–860 m in the Ya'an-Tianquan area (Figure 1b), and that the difference in SAR between the two flanks varies by a factor of ~2. The interpretation of seismic profiles conducted by Jia et al. (2006) in the Longmen Mountain and the Sichuan Basin indicated that deflection deformation occurred on the western margin of the Yangtze Craton during the Late Cretaceous–Paleocene, forming a subsidence center in the Ya'an area within the Longmenshan foreland basin. This supports the conclusion that the thickness of the Guankou Formation is greater in the Ya'an area than in the Tianquan area. Combining the magnetic polarity column correlations and the SAR characteristics for both flanks of the fold, we conclude that the two intervals of rapid sediment deposition (during 128–124.1 and 68.8–64.1 Ma) identified in this study are reliably defined.

The two intervals of increased SAR identified in this study (128.0–124.1 and 68.8–64.1 Ma) coincide with an increase in coarse-grained sediments. However, an explanation is needed for the occurrence of a set of

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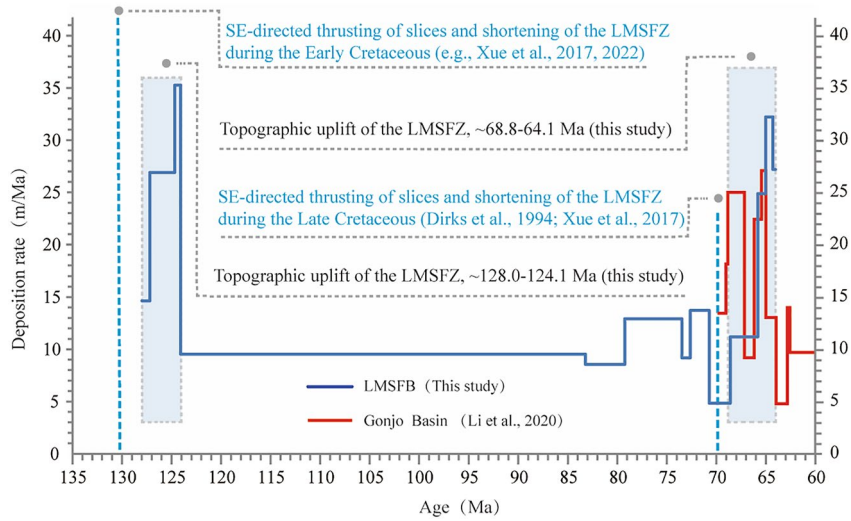


Figure 9. Relationship between rapid sediment deposition in the Longmenshan foreland basin and the Gonjo Basin (S. H. Li, van Hinsbergen, et al., 2020), and strong activity of the LMSFZ. LMSFZ, Longmenshan Fault Zone; LMSFB, Longmenshan foreland basin.

conglomerate deposits at the base of the Guankou Formation which are not characterized by any change in SAR (Figure 9). We suggest that this phenomenon is unlikely to be related to any interruption of sediment deposition or to local uplift within the basin. The main reasons for this are as follows: (a) We sampled a continuous section and observed no evidence of sedimentary discontinuities or interruptions during field investigations. (b) The gravels in the conglomerate layer at the base of the Guankou Formation are well rounded and the lithologies of most of the gravels are strongly deformed metamorphic and volcanic rocks, which indicates that these gravels were deposited after remote transportation rather than being the products of the erosion of the lower strata (Figure 2e). Additionally, there are several sandstone interbeds (with thicknesses of ~0.3–1 m) within the conglomerate layer. (c) At Feixianguan (Figure 1c) there are no conglomerate deposits at the base of the Guankou Formation (Shen, 2018). If the conglomerate at the base of the Guankou Formation was the result of local uplift, conglomerate should also be present at the base of the Guankou Formation in the Feixianguan section, as they are only ~10 km apart. Therefore, we consider it unlikely that the conglomerates at the base the Guankou Formation are associated with a sedimentary hiatus or local uplift within the basin.

The SAR of the R7 polarity interval (the base of the Guankou Formation) did not change significantly, likely because the corresponding stratigraphic thickness of R7 (with the SAR of ~8.5 m/Myr) is ~30 m. The conglomerate intercalated with sandstone deposits at the base comprises only ~10 m, and the upward ~20 m mainly comprise mudstone intercalated with siltstone deposits (likely with a relatively low SAR) (Figure 7), which may result in no major change in the average SAR of the R7 polarity interval. Assuming the SAR of the conglomerate intercalated with sandstone (~10-m-thick) at the base of the R7 polarity interval was very high (the deposition time can be ignored), a minimum SAR of 5.7 m/Myr can be estimated for the upward ~20 m of sandstone intercalated with siltstone in the R7 polarity interval. Even if the SAR of the R7 polarity interval was 5.7 m/Myr (9.5 m/Myr for N7), this has no effect on the inference of the occurrence of two stages of rapid deposition (during 128.0–124.1 and 68.8–64.1 Ma). Also, since the conglomerate intercalated with sandstone at the base of the R7 polarity interval is only ~10-m thick, even if it was deposited rapidly, it would only have responded to very short-term SAR fluctuations (assuming an SAR of 30 m/Myr and a deposition time of only ~0.3 Myr). However, it is unlikely that the very brief fluctuations in SAR were the result of the uplift of the Longmen Mountains and they are more likely to be the result of brief fluctuations of the regional climate. Therefore, although conglomerate intercalated with sandstone is present at the base of the R7 polarity interval, it does not affect our inference of two stages of rapid deposition (128.0–124.1 and 68.8–64.1 Ma).

To sum up, our magnetostratigraphic results reveal the occurrence of two rapid depositional events (during 128.0–124.1 and 68.8–64.1 Ma) and one slow depositional event (124.1–68.8 Ma) in the Longmenshan foreland basin from 128 Ma to 64.1 Ma.

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4.3. Cause of the Two Rapid Depositional Events

Previous studies have proposed that the SAR in terrestrial basins depends on the rate of denudation of the source area (Hovius, 1998). Denudation rates are affected by several factors, including the average elevation of the watershed, precipitation, runoff, slope, average watershed temperature, and the temperature range (Harrison, 1994; Hovius, 1998; Willenbring et al., 2013). These factors are related to the topography (the product of tectonic activity) and the climatic conditions which influence sediment transport (Chang et al., 2015; Charreau et al., 2006; Molnar & England, 1990).

Tectonic analysis indicates that the LMSFZ was strongly compressed by the Songpan-Ganzi block from NW to SE during the Early Cretaceous and Late Cretaceous–Paleocene (Dirks et al., 1994; Y. Li et al., 2016; Z. Xue et al., 2017; Z. H. Xue et al., 2022), which led to the SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ (Dirks et al., 1994; Y. Li et al., 2016; Z. Xue et al., 2017; Z. H. Xue et al., 2022). Thermochronological studies show that the LMSFZ experienced rapid cooling during the Early Cretaceous (e.g., Arne et al., 1997; Z. W. Li et al., 2012; Kirby et al., 2002; Xu et al., 2007) and the Late Cretaceous (Tian et al., 2016; Wallis et al., 2003). Balanced geologic cross-sections analysis suggests that tectonic shortening (SE-directed thrusting of basement-slices) in the LMSFZ is the main driver of topographic uplift in the Longmen Mountains (Hubbard & Shaw, 2009; Hubbard et al., 2010), and that a small horizontal shortening (1.6%–2.7%) would result in uplift of up to 37% (Hubbard et al., 2010).

Source analysis has demonstrated that the provenance of the materials in the basin is mainly the Longmen Mountains (Y. Li et al., 2016; Peng et al., 2000; Richardson et al., 2008). As the topography of the Longmen Mountains rises, the rate of denudation will increase (Charreau et al., 2006; Hovius et al., 1998). Therefore, the SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ during the Early Cretaceous and Late Cretaceous–Paleocene would have led to increased SARs within the basin (Charreau et al., 2006; Hovius et al., 1998). Our results constrain the increase in SARs within the basin to the intervals of 128.0–124.1 and 68.8–64.1 Ma. The earlier tectonic rapid deposition (128.0–124.1 Ma) corresponds to SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ during the Early Cretaceous, and the later phase (68.8–64.1 Ma) corresponds to SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ during the Late Cretaceous–Paleocene (e.g., Dirks et al., 1994; Jia et al., 2006; Y. Li et al., 2016; Z. Xue et al., 2017; Z. H. Xue et al., 2022).

Climate change is an additional factor that may effect changes in the SAR within the basin. We assume that the two phases of rapid deposition were climate-dominated, which implies significant climatic changes within the basin during the Cretaceous. However, the results of paleoclimatic studies conducted within the basin by J. Li, He, et al. (2018) and J. Li, Huang, et al. (2020) do not suggest significant climate changes within the basin at 128–124.1 Ma and 68.8–64.1 Ma. Although the paleoclimate results of J. Li, He, et al. (2018) and J. Li, Huang, et al. (2020) are of relatively low resolution, they still provide a reliable record of regional climate change. In addition, in the tectonically-controlled Longmenshan foreland basin, the SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ will result in topographic uplift of the Longmen Mountain (Hubbard et al., 2010; Hubbard & Shaw, 2009) and increased SARs within the basin (Charreau et al., 2006; Hovius et al., 1998). The strong correlation between the two phases of rapid deposition and the SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ suggests that the uplift of Longmen Mountain caused by tectonic activity was the dominant factor in the two phases of rapid deposition (128–124.1 and 68.8–64.1 Ma).

4.4. Tectonic Significance of Cretaceous Uplift in the Longmen Mountains

On the eastern side of the Tibetan Plateau, the Longmen Mountains rise 6,000 m relative to the Sichuan Basin, exhibiting greater relief than anywhere else on the plateau (Hubbard & Shaw, 2009). Several studies have suggested that the LMSFZ crust may have been relatively thick before the collision between India and Asia (e.g., Wallis et al., 2003; Xu et al., 2007; Z. Xue et al., 2017; Z. H. Xue et al., 2022), and that the Cenozoic thickening of the LMSFZ crust may have been be less significant than is often suggested (Z. Xue et al., 2017). Our results constrain the uplift of the Longmen Mountains during the Cretaceous–Early Paleogene to the intervals of 128–124.1 and 68.8–64.1 Ma, supporting the view that the LMSFZ crust thickened before the collision between India and Asia.

Numerous studies have shown that the subduction and closure of the Bangong-Nujiang Ocean during the Jurassic–Cretaceous led to a significant shortening and thickening of the crust in west and central Tibet prior to

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the India-Asia collision (e.g., Y. L. Li et al., 2015; Kapp et al., 2003, 2005, 2007; Murphy et al., 1997; Volkmer et al., 2007, 2014; Xu et al., 2016). Field studies have provided shortening estimates of >50% in the west and central areas of the Lhasa terrane during the Cretaceous–Paleocene (Kapp et al., 2003, 2005; Murphy et al., 1997). Tectonic analysis shows that SE-directed thrusting of basement-slices and tectonic shortening of the LMSFZ during early Cretaceous was caused by the SE-directed compression of the Songpan-Ganzi block (Dirks et al., 1994; Y. Li et al., 2016; Z. Xue et al., 2017; Z. H. Xue et al., 2022). Although our results can only constrain the closure time of the Bangong-Nujiang Ocean to a broad range (Jurassic–Cretaceous), the SE-directed compression of the LMSFZ by the Songpan-Ganzi block and the uplift of Longmen Mountains during the Early Cretaceous imply that the stress field generated by the subduction and closure of the Bangong-Nujiang Ocean had already affected the eastern edge of the Tibet in the Early Cretaceous (at ∼128 Ma).

5. Conclusions

We have conducted a detailed magnetostratigraphic study of the Cretaceous Jiaguan Formation and the Guankou Formation, in the Jinmenguan section at the southwestern margin of the Longmenshan foreland basin. The resulting magnetostratigraphy enables us to precisely constrain the stratigraphic age of the Jiaguan Formation and to determine SARs for the Jinmenguan section. We identify two periods of rapid deposition (at 128.0–124.1 and 68.8–64.1 Ma) that constrain two phases of tectonic uplift in the Longmen Mountains during the Cretaceous–Early Cenozoic. Our results lead to the following major conclusions. (a) The stratigraphic age of the Jiaguan Formation has the range of 128.0–83.6 Ma. (b) Two phases of tectonic uplift occurred in the LMSFZ during the Cretaceous–Early Cenozoic (128.0–124.1, 68.8–64.1 Ma), and thus our results support the view that the crust of the Longmen Mountains thickened prior to the India-Asia collision. (c) The stress field generated by the subduction and closure of the Bangong-Nujiang Ocean had already been transferred to the eastern edge of the Tibet in the Cretaceous (at ∼128 Ma).

Data Availability Statement

The data in this study are available for download through share at https://figshare.com/s/947f690d869630db8cde.

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